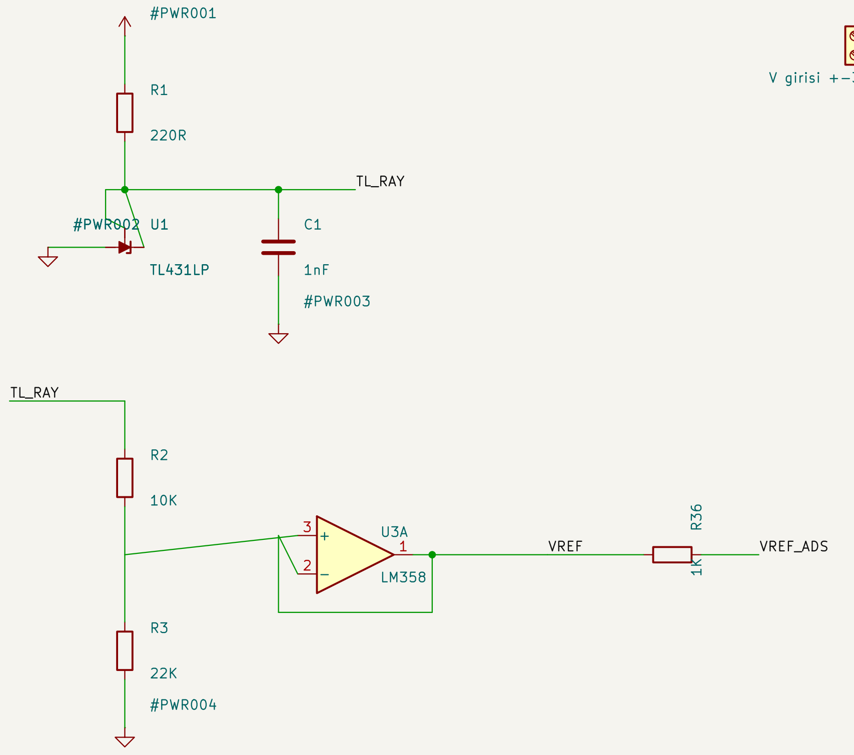
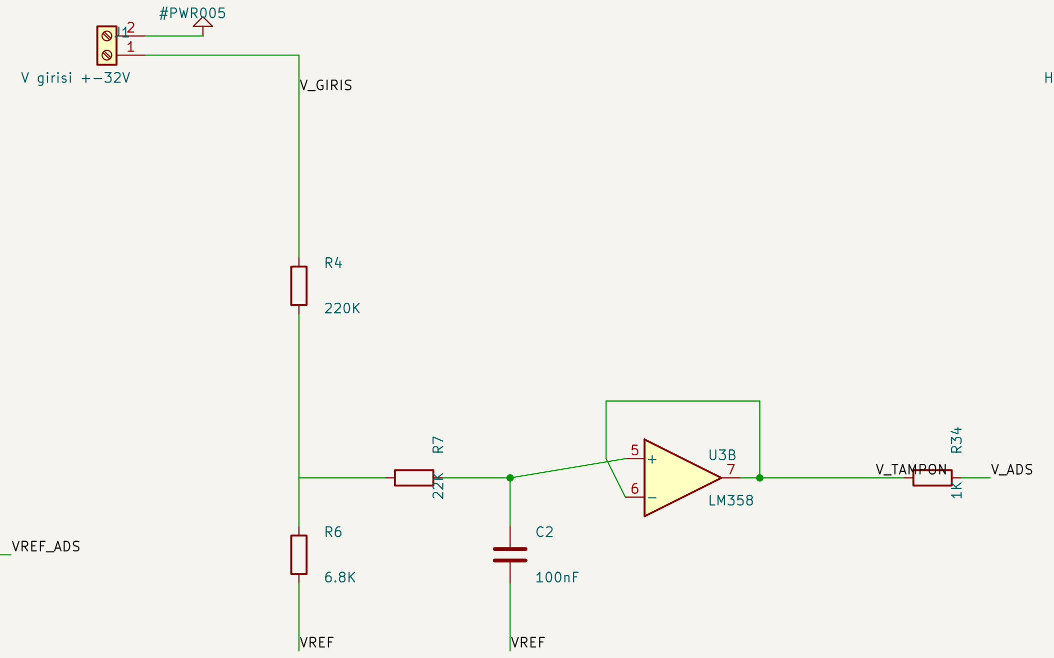
BLOK 1 — TL431 rayi 2.495 V + Vref 1.7153 V (TAMPONLU)



Vref TAMPONSUZ OLAMAZ: iki bolucunun akimi (~290 uA) 6.9 kohm'luk bolucu Thevenin'inden aksa 2 V'a varan kayma yapardi.

Vref hatasi girise vurulmus SABIT ofsettir (N'den bagimsiz), kazanc hatasi degil — sifir kalibrasyonu siler.

BLOK 2 — NORMAL gerilim +-32.4 V (220K / 6.8K) -> ADS #2 AIN0



Menzil Vref kadar YUKARI kaymistir: -32.44 .. +35.87 V.

Simetrik menzil +-32.44 V @ PGA +-1.024, adim 1.042 mV.

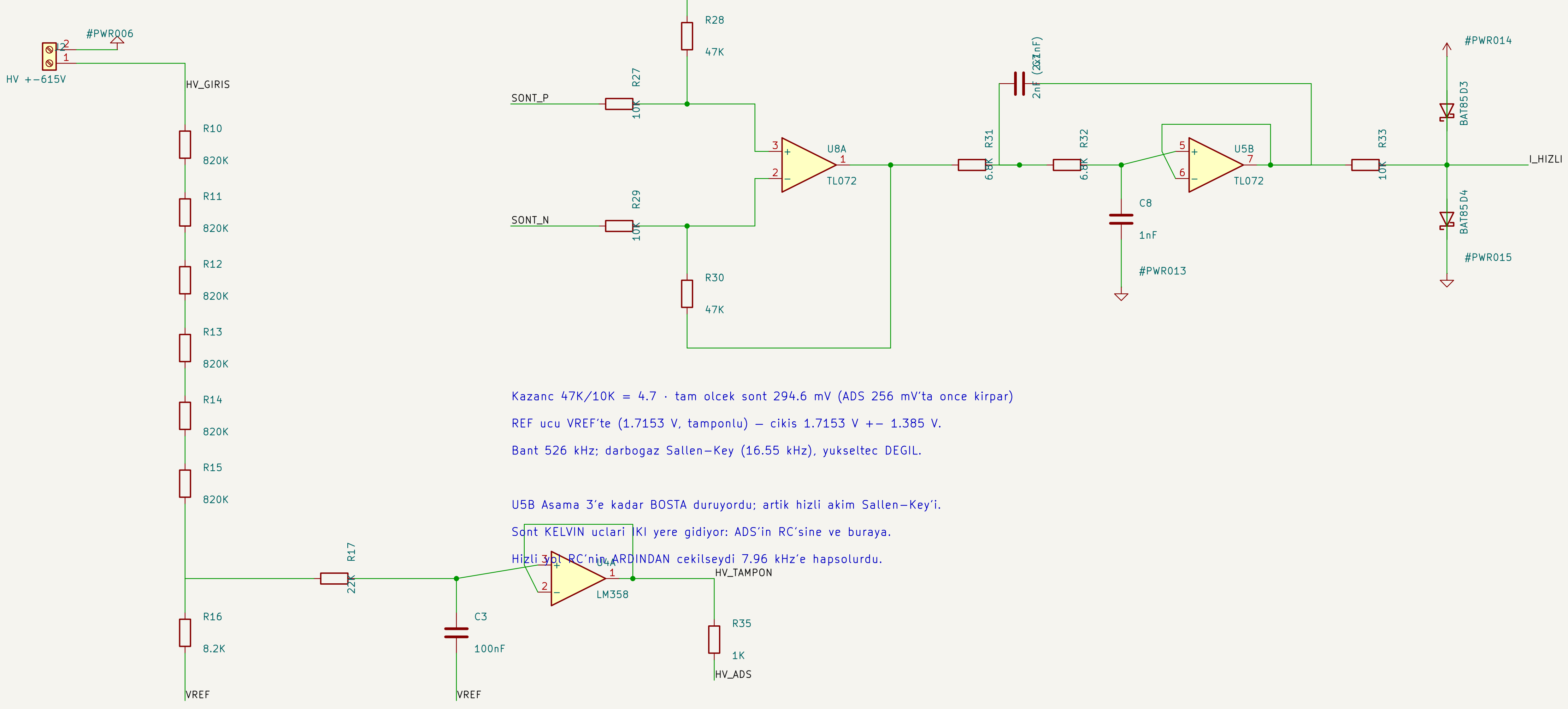
Bolucunun ALT ucu GND'ye degil VREF'e gidiyor: fark = (Vin-Vref)/N,

ADS diferansiyel okudugu icin ISARETLI -> negatif kendiliginden geliyor.

TAMPON ZORUNLU: tamponsuz bolucude ADS'in giris empedansi PGA ile degistigi icin oto-kademe %0.67 puan kazanc sicramasi yapiyor (4.14).

BLOK 3 — YUKSEK gerilim +-613.7 V (6x820K / 8.2K) -> ADS #2 AIN2

BLOK 8 — Hizli akim yolu: fark yukselteci -> Sallen-Key -> GPIO5



Kazanc 47K/10K = 4.7 · tam olcek sont 294.6 mV (ADS 256 mV'ta once kirpar)

REF ucu VREF'te (1.7153 V, tamponlu) — cikis 1.7153 V +- 1.385 V.

Bant 526 kHz; darbogaz Sallen-Key (16.55 kHz), yukseltec DEGIL.

U5B Asama 3'e kadar BOSTA duruyordu; artik hizli akim Sallen-Key'i.

Sont KELVIN uclari IKI yere gidiyor: ADS'in RC'sine ve buraya.

Hizli yol RC'nin ARDINDAN cekilseydi 7.96 kHz'e hapsolurdu.

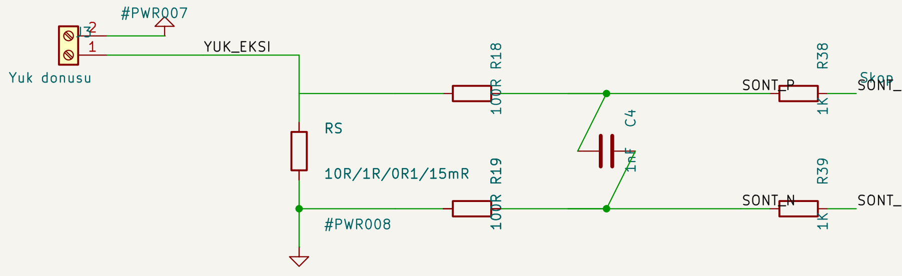
!!! 615 V OLDURUR. Delikli plakette bu 6 direnc DELIK ATLAYARAK, aralarinda bosluk birakilarak konmali — 2.54 mm adim YETMEZ.

6 adet 820K: 1/4W metal filmin azami CALISMA gerilimi 200 V (Yageo MFR).

615 V'ta direnc basina 102 V = sinirin %51'i. Tedarikcinin METAL FILM hatti 820K'da bitiyor; 6x820K / 8.2K N'yi TAM 601.0'da tutuyor.

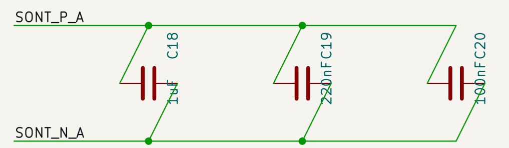
Giris empedansi 4.93 Mohm (Asama 2'de 106.8 kohm idi).

BLOK 4 — Sont + KELVIN + diferansiyel RC -> ADS #1 AIN0/AIN1 (CIFT YONLU)



Asama 2'de firmware negatif akimi SIFIRA KIRPIYORDU. Donanim zaten diferansiyel ve isaretli; kirpma kaldirildi (olcum3.h). Guc elektroniginde bobin akimi ters doner — tek yonlu olcum yanlis cevaptir.

B16 — ADS akim girisi ortusme suzgeci: C18+C19+C20 = 1.32 uF (gerilim kanaliyla tau eslesmesi)

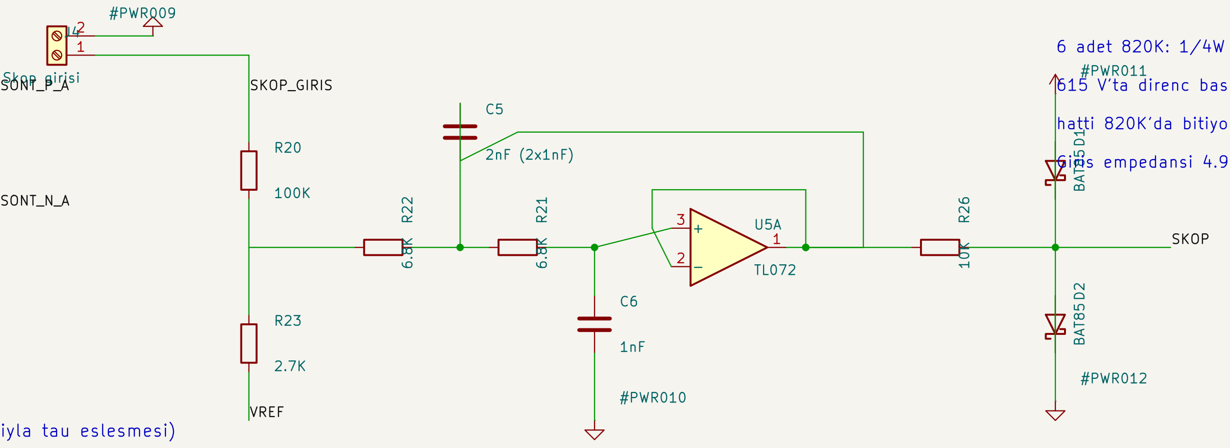


Toplam 1.320 uF; ideal 1.308 uF (%1.0 uste).

Kesim 54.8 Hz — gerilim kanalinin 55.7 Hz'i ile eslesiyor ve 430 Hz Nyquist'i koruyor.

!!! ADS pinlerine YAKIN, KISA bacakli.

BLOK 5 — Osiloskop CIFT YONLU: -65..+45 V -> Sallen-Key 16.55 kHz -> BAT85 -> GPIO4



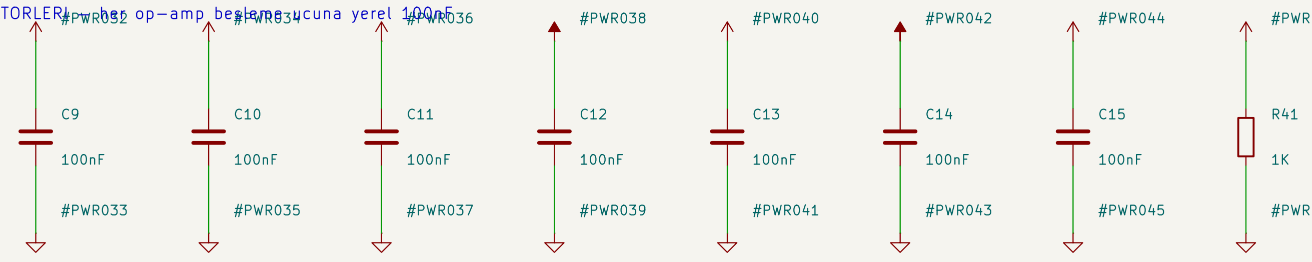
BAT85 (DO-34 EKSENEL, delikli) — BAT54 SOT-23 oldugu icin secilmedi.

1N4148 OLMAZ: B2 olctu, ariza akiminin %25'i ADS'in ic ESD diyoduna gidiyor. BAT85 ile pin -0.268 V'ta kaliyor, ESD payi ~%0.

Seri direnc 2.7K: 1K olsaydi +-12 V arizada 11.3 mA — sinir 10 mA.

Sallen-Key f0 16.55 kHz, Q 0.707. Eski tek kutuplu RC 25 kHz'teydi — Nyquist'in (20.83 kHz) USTUNDE, yani hicbir sey suzmuyordu (4.8).

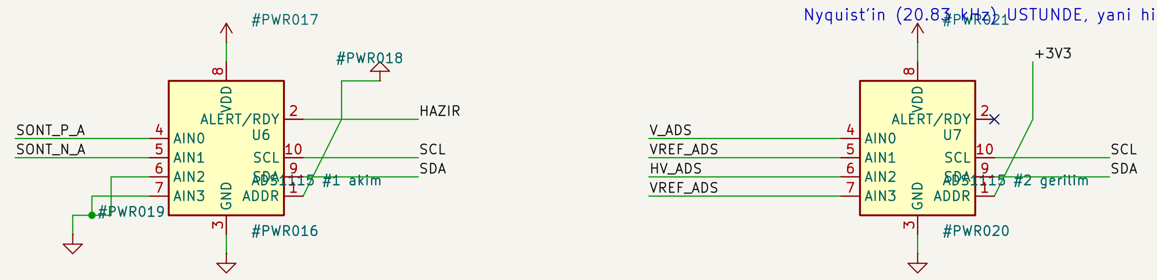
AYIRMA KONDANSATORLERI — her op-amp besleme ucuna yerel 100nF

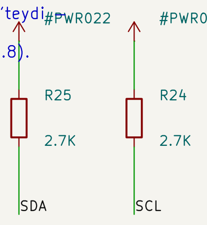


C9 U3 · C10 U4 · C11/C12 U5 · C13/C14 U8 · C15 ADS rayi

R41 (B18/F12): 3V3 bosaltma. Beslemesiz kartta kelepce akimina yol verir; ray 3.58 V yerine 1.67 V'ta kalir ve TL431'e bagimli olmaz.

BLOK 6 — ADS1115 x2 · I2C pull-up 2.7K (4.7K DEGIL)

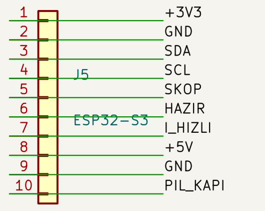




4.7K iki modulde 2.42 kohm -> t_r 308 ns; Fast-mode siniri 300 ns.

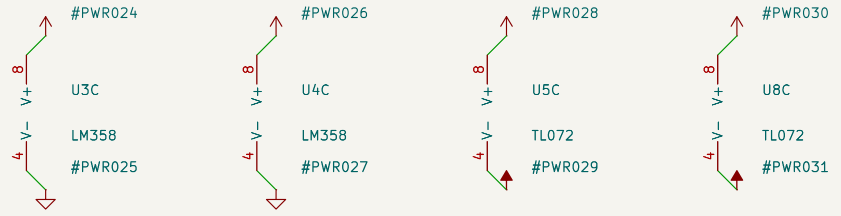
2.7K hem 2 hem 3 modulde geciyor (223 / 190 ns), sink 2.21 mA < 3 mA.

BLOK 7 — ESP32-S3 N16R8

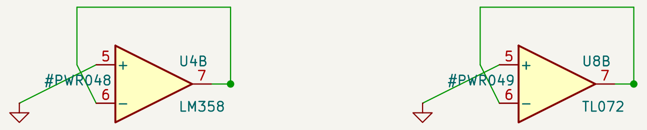


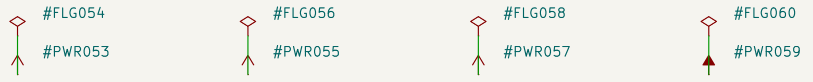
GPIO8 SDA · GPIO9 SCL · GPIO4 SKOP (ADC1_CH3) · GPIO5 I_HIZLI (CH4) · GPIO7 HAZIR

OP-AMP BESLEMELERI

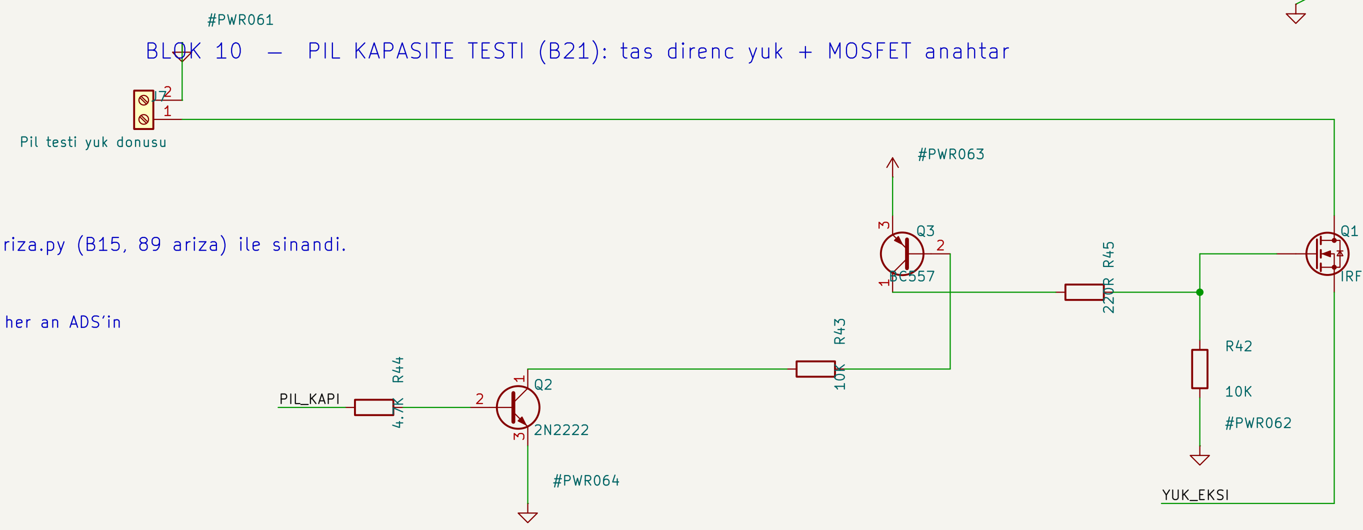


KULLANILMAYAN KESITLER — izleyici olarak baglandi





BLOK 10 — PIL KAPASITE TESTI (B21): tas direnc yuk + MOSFET anahtar

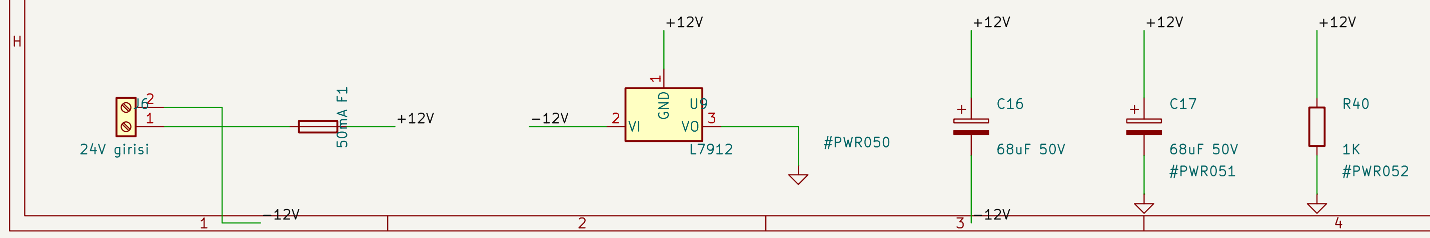


Asama 3 — CIFT YONLU on uc. Tum degerler uretim/tasarim3.py (89 kural), uretim/sim3_giris.py (30 SPICE) ve uretim/sim3_ariza.py (B15, 89 ariza) ile sinandi.

KART IZOLE DEGIL. 615 V kanali sebeke referansli bir devreye BAGLANMAZ — USB uzerinden bilgisayara sebeke tasir.

B15 KORUMALARI: R34/R35/R36 (1K) tampon cikislari ile ADS pinleri arasinda; R38/R39 (1K) sont kolunda. Bunlar OLMADAN tampon doydugu her an ADS'in mutlak maksimumu (3.6 V) asiliyordu ve sont arizasi ADS'i olduruyordu. C1 = 1nF (100nF TL431'i osile ettiriyordu — TI SLVA482A).

BLOK 9 — +-12 V RAYI (B11): 24 V -> 7912 orta nokta regulatoru



7912: GND pini +12V'ta, VI -12V'ta, VO kart GND'sinde. Orta noktayi TANIMLAYAN parca bu.

Kapi GND'ye CEKILI (R42): ESP32 olurse/reset atarsa MOSFET KAPANIR ve pil bosalmayi durdurur. TERS KURMAYIN (kapiyi +12 V'a cekmeyin).

Pil gerilimi <= 38.5 V (IRFZ44N Vdss 55 V, %70 pay) — 48 V paket OLMAZ.

Yuk J7'ye baglanir. J3'e baglanirsa MOSFET BAYPAS olur ve KESME CALISMAZ — firmware test basinda bunu denetleyip testi reddediyor.

tasarim3.py + sim3_giris.py + netlist3_dogrula.py ile dogrulandi
Voltmetre +-31.1 V / 950 uV ve +-615.4 V / 18.8 mV · Ampermetre CIFT YONLU · Osiloskop Sallen
Elektronik Calisma Alani
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Title: Olcum Karti — Asama 3 (CIFT YONLU on uc, +-615 V)

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